RAJALAKSHMI ENGINEERING COLLEGE

Dhanusheya G 251801053-2025-AIDS _

C2

Started on	Thursday, 17 September 2026, 8:24 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:24 AM
Time taken	33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

#define MAX_LEN 1005

int dp[MAX_LEN][MAX_LEN];

int max(int a, int b) {
    return (a > b) ? a : b;
}

int main() {
    char s1[MAX_LEN];
    char s2[MAX_LEN];

    // Read the two strings silently from standard input
    if (scanf("%s", s1) != 1) return 0;
    if (scanf("%s", s2) != 1) return 0;

    int m = strlen(s1);
    int n = strlen(s2);

    // Build the DP table from the bottom up
    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0) {
                dp[i][j] = 0; // Base case: empty strings have 0 common subsequence length
            } else if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }

    // The bottom-right cell contains the length of the longest common subsequence
```

```
36      // The bottom-right cell contains the length of the longest common subsequence
37      printf("%d\n", dp[m][n]);
38
39      return 0;
40  }
41
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course